

RESEARCH EXPERIENCE

Computational Imaging Lab | VISITING STUDENT MAR 2025 – NOW
UC Berkeley, EECS

- Working with **Prof. Laura Waller**, designing a computational spectrometer with information-driven deep learning frameworks and building a setup for optical diffraction tomography.
- Ongoing Research:** Info-driven Optimization, Multi-wave ODT

Optical Imaging System Lab | PH.D CANDIDATE MAR 2019 – NOW
Yonsei University, Electronic and Electric Engineering

- Conducted research under the supervision of **Prof. Seung Ah Lee** in the Optical Imaging System Lab(now Imaging Intelligence Lab), focusing on lensless imaging techniques and computational imaging methods.
- Ongoing Research:** 3D Depth imaging, Task-Optimized PSF Design

RESEARCH PROJECTS

Lensless 3D Depth Imaging | 2023 - Now
- Keywords: Depth Imaging, PSF Engineering, Wave optics, SLM

- Design a lensless camera for 3D imaging, enriching depth resolution with optimized phase masks forming multiple depth-dependent point spread functions using a spatial light modulator.

Lensless Snapshot Hyperspectral Imaging | 2021 - 2023
- Keywords: Hyperspectral Imaging, Linear variable filter, Calibration

- Build a snapshot lensless imaging system that utilizes a linear variable filter and a phase mask to encode spectral information, enabling recovery of hyperspectral image stacks from a single measurement.

Lensless High Speed Eye Tracking | 2019 - 2021
- Keywords: Eye tracking, Wearable device, Rolling-shutter

- Design and construction of a goggle-type eye tracker using a low-cost and high-speed lensless camera with rolling shutter image sensor for monitoring eye movements in neurodegenerative diseases.

PUBLICATIONS

JOURNAL

- Kim, T., Lee, K. C., Baek, N., Chae, H., & Lee, S. A. (2023). **Aperture-encoded snapshot hyperspectral imaging with a lensless camera**. APL Photonics, 8(6).
- Baek, N., Lee, Y., Kim, T., Jung, J., & Lee, S. A. (2022). **Lensless polarization camera for single-shot full-Stokes imaging**. APL Photonics, 7(11).
- Lee, Y., Chae, H., Lee, K. C., Baek, N., Kim, T., Jung, J., & Lee, S. A. (2022). **Fabrication of integrated lensless cameras via UV-imprint lithography**. IEEE Photonics Journal, 14(2), 1-8.

PROCEEDING

- Kim, T., Lee, Y., Jung, J., Lee, K. C., & Lee, S. A. (2024, August). **3D Lensless Camera for extended depth range with multiple point spread functions**. In Conference on Lasers and Electro-Optics/Pacific Rim (p. We3G_3). Optica Publishing Group.
- Kim, T., Lee, K. C., Lee, K., Baek, N., Jung, J., Kim, E., ... & Lee, S. A. (2024, February). **High-speed lensless eye tracker for microsaccade measurement**. In SPIE Advanced Biophotonics Conference (SPIE ABC 2023) (Vol. 13076, pp. 38-44). SPIE.

RECENT CONFERENCE ACTIVITY

- Miniaturized Computational Spectrometer with Information-driven Design (Poster)** (Photonics West 2026)
- Privacy-Preserving Imaging With Lensless Cameras Using Shift-Variant Point Spread Functions (Oral)** (COSI 2025)
- Task-Optimized Lensless Imaging With Trainable Phase Masks for Compact Spatial Encoding (Oral)** (COSI 2025)
- Quantitative Fluorescence Imaging for Detection of Dental Plaque with Lensless Camera (Poster)** (ABC 2024)
- 3D Lensless Camera for Extended Depth Range with Multiple Point Spread Functions (Oral)** (CLEO-PR 2024)

EDUCATION

YONSEI UNIVERSITY
MS & PH.D OF ENGINEERING IN SCHOOL OF ELECTRONIC AND ELECTRICAL ENGINEERING
Expected Sep 2026 | Republic of Korea, Seoul

YONSEI UNIVERSITY
BACHELOR OF ENGINEERING IN SCHOOL OF ELECTRONIC AND ELECTRICAL ENGINEERING
Feb 2021 | Republic of Korea, Seoul

HONORS

AWARDS

- 2024 ABC Best Poster Award**, Optical Society of Korea (2024)
- 2023 SPIE-ABC Young Investigator Award**, Optical Society of Korea (2023)
- Outstanding Presentation Award**, Capstone Design Project, (2019)

PRESS

- AIP Scilight**, “Improving hyperspectral imaging using a lensless camera”, 2023

GRANTS

- Selected Participant**, BK21(Brain Korea 21) International Joint Training Program, National Research Foundation of Korea (NRF), 2025
- Undergraduate Research Fellow**, Undergraduate Research Program, Korea Science and Creativity Foundation (KOSAC), 2019

SOCIETY

- Regular Member**, Yonsei EEHS(Honor Society)

PATENTS

- Apparatus and method for measuring eye movement** - Lee, S. A., Kim, T. Y., Lee, K. W., Baek, N. K., & Jung, J. W. (2024). U.S. Patent Application No. 18/052,192.
- Methods for manufacturing phase masks and lens-less camera module** - Lee, S. A., Chae, H. S., Lee, Y. J., Lee, K. C., Baek, N. K., Kim, T. Y., & Jung, J. W. (2023). U.S. Patent Application No. 18/168,887.

SKILLS

TECHNICAL
Python(Pytorch, OpenCV) • MATLAB • Fusion 360

LANGUAGES
Korean (Native) • English (Intermediate, OPIc IM)

ADDITIONAL INFO

For Other Projects / Publications / Patents / Conference Activities

